

Topics in Software Engineering– Software Reengineering

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Course Objectives & Course Outline



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: **CSE327**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Software Re-Engineering**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Catalogue Description:

This course provides the concepts and application of re-engineering techniques. Topics include: Introduction to Software Re-Engineering; Software Evolution & Legacy Systems; Re-Engineering Techniques; Refactoring & Applications; Re-Architecting; Big Rewrite; Project Improvement & Automating the Development Environment; Object Oriented Re-Engineering Patterns; Quality Issues; Tool Support; Challenges & Stakeholder Aspiration; Software Maintenance; and Re-Engineering Economics.

Unit wise Major Topics:

Unit	Topic	No. of Teaching Hours
1.	Software Re-Engineering: Introduction, Evolution, and Legacy Systems.	3
2.	Techniques of Re-Engineering: Reverse Engineering, Restructuring, and Forward Propagation.	7.5
3.	Refactoring & Applications; Re-Architecting; and Big Rewrite.	6
4.	Project Improvement & Automating the Development Environment.	6
5.	Object Oriented Re-engineering Patterns.	7.5
6.	Quality Issues in Re-Engineering Processes.	4.5
7.	Tool Support for Re-engineering; and Challenges & Stakeholder Aspiration.	6
8.	Software Maintenance; and Re-engineering Economics.	4.5
Total Contact Hours		45

Student Outcomes (SOs)

S.#	Description
1	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
2	Identify, formulate, research literature, and solve <i>complex</i> computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
3	Design and evaluate solutions for <i>complex</i> computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to <i>complex</i> computing activities, with an understanding of the limitations.
5	Function effectively as an individual and as a member or leader in diverse teams and in multi disciplinary settings.

Mapping of CLOs and SOs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO-1	1	Recognize the concepts of software re-engineering.	<i>Understanding</i>	1
CLO-2	2-4	Apply re-engineering techniques to maintain and modify software systems.	<i>Applying</i>	2,4
CLO-3	5	Apply Object oriented re-engineering patterns to maintenance problems.	<i>Applying</i>	2,4
CLO-4	6	Illustrate quality and tools support issues and challenges.	<i>Applying</i>	2,4
CLO-5	3-8	Reconstruct low-level source code structure of a legacy system by applying refactoring strategies and	<i>Creating</i>	2,4,9

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-
Assignments	-	Assignment 1	Assignment 2	Assignment 3	Assignment 4
Mid Term Exam	Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-
Final Term Exam	Final Term Exam				

Text and Reference Books**Textbook:**

1. Re-Engineering Legacy Software, David Lorge Parnas, Chris Birchall, Manning publications, 2016.

Reference Book:

1. Re-Engineering, Priyadarshi Tripathy, Kshirasagar Naik, John Wiley & Sons, Inc, 2015.

- **Attendance Policy:** Every student must attend 80% of the lectures delivered in this course and 80% of the practical/laboratory work prescribed for the respective courses. The students falling short of required percentage of attendance of lectures/seminars/practical/laboratory work, etc., shall not be allowed to appear in the terminal examination of this course and shall be treated as having failed this course.

· **Course Assessment:**

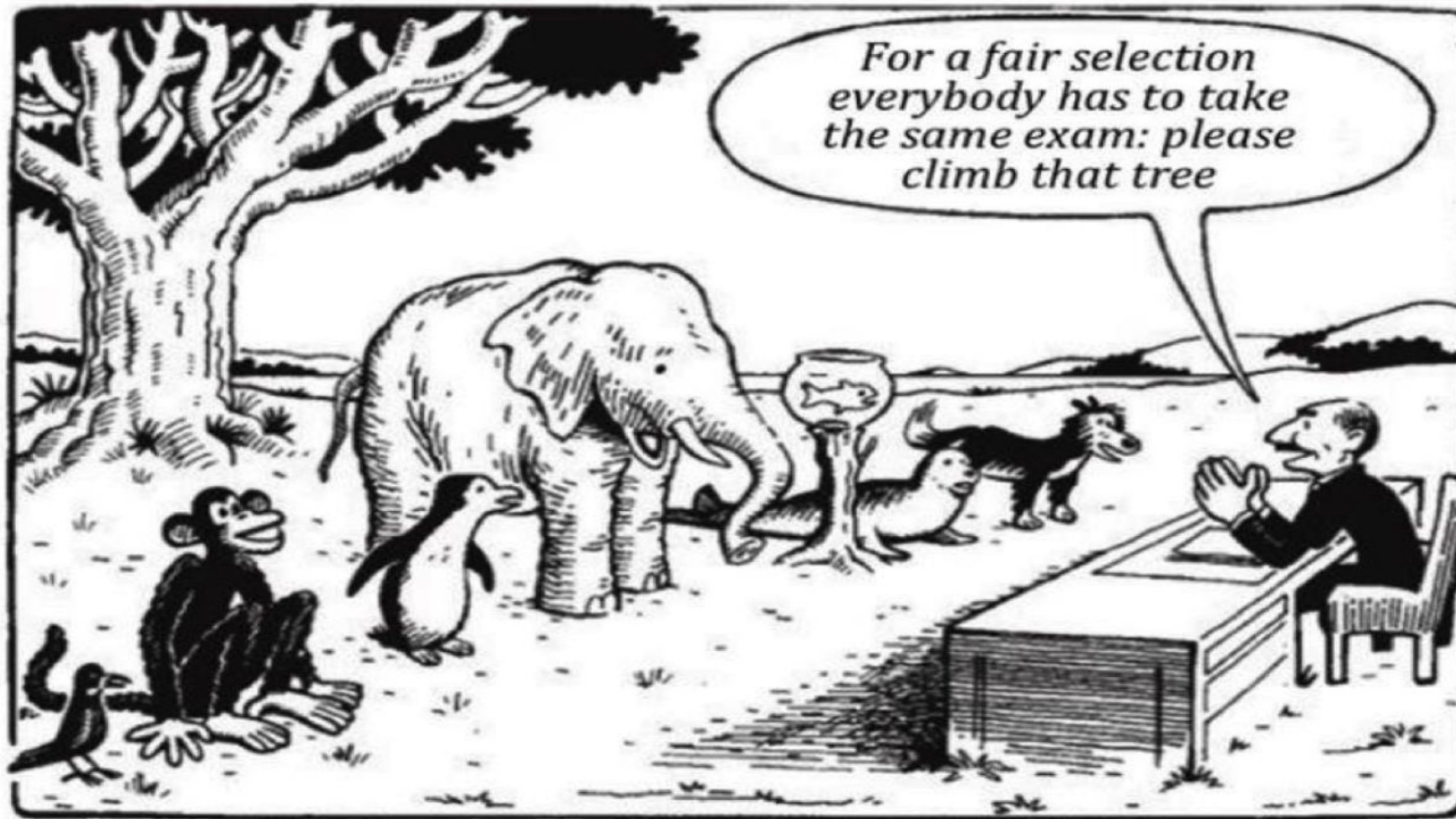
	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Total
Theory (T)	15	10	25	50	100
Final Marks (T)	100				

- **Grading Policy:** The minimum pass marks for each course shall be 50%. Students obtaining less than 50% marks in any course shall be deemed to have failed in that course. The correspondence between letter grades, credit points, and percentage marks at CUI shall be as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D	F
Marks	90 - 100	85 - 89	80 - 84	75 - 79	70 - 74	65 - 69	60 - 64	55 - 59	50 - 54	<50
Cr. Point	4.0	3.7	3.3	3.0	2.7	2.3	2.0	1.7	1.3	0.0

OBE Based Learning

Outcome Based Education(OBE)



Our Education System

"Everybody is a genius. But if you judge a fish by its ability to climb a tree, it will live its whole life believing that it is stupid."

- Albert Einstein

OBE Component

OBE
(Education)

What the student
should be able to do?

OBC
(Curriculum)

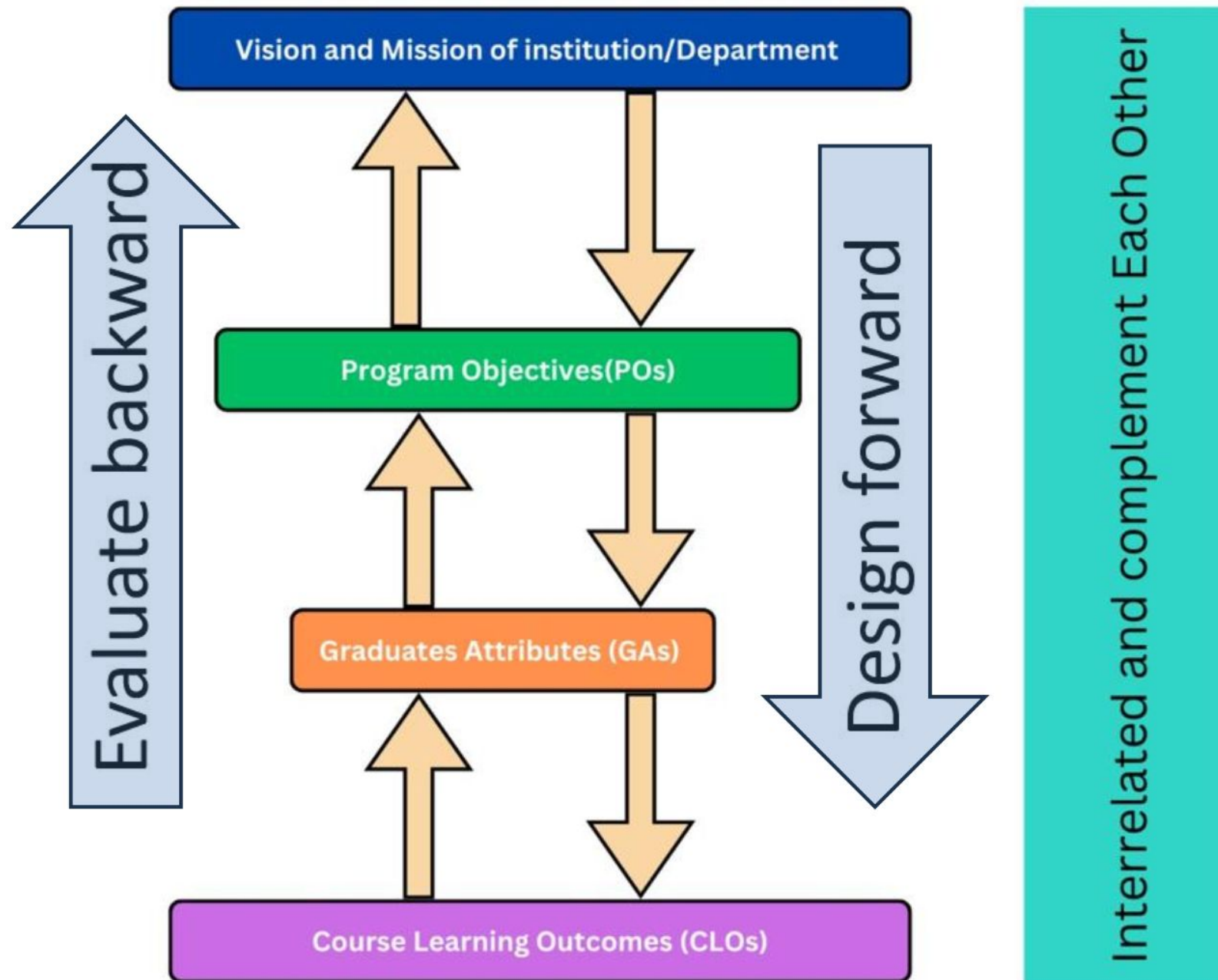
How to make the
student achieve the
outcome?

Attributes

OBLT
(Learning & Teaching)

How to measure what the
student has achieved?

OBA
(Assessment)



Assessments levels, methods, & frequency

- **Course Learning Outcomes (Every semester)**

- ▣ Direct Assessments - Assignments, Quizzes, Mid and Final terms, Course projects, and Capstone project

- **Graduate Attributes (Once a year)**

- ▣ Direct Assessments - Course level assessments
- ▣ Indirect Assessments - Exit Survey

- **Program Objectives (After every two years)**

- ▣ Indirect Assessments – Alumni and Employee Survey

Program Objectives

PO approved from Academic Council(AC)

- 
- *Inculcate in-depth knowledge, analytical skills, and creativity in the computing domain.*
 - *Attain the ability to adapt in an evolving technological environments, assimilate new information with a strong focus on entrepreneurship.*
 - *Instill moral and ethical values, along with the ability to communicate effectively with computing community.*
 - *Train graduates to contribute towards knowledge economy and socio-economic growth of the country.*

Bloom's Taxonomy (Revised)

Action Verbs

Identify Define
List Label Match
Name Select Recall
Recognize Repeat
State

Remember

Recognize or recall knowledge from memory

Quizzes with multiple choice / T/F/ Fill-in-blank questions

Classify Compare
Describe Distinguish
Discuss Explain
Illustrate
Select Summarize
Translate
Rank Rate

Understand

Comprehend meaning from instructional messages

Short-answer question
Discussions
Concept map
Comparison chart

Apply Calculate
Compute Develop
Execute
Graph Relate Use
Operate
Organize Practice
Implement Solve

Apply

Carry out procedures in a given situation

Problem Solving
Demonstrations
Sketches
Simulations
Case studies
Role play

Analyze
Inquiry
Differentiate
Organize
Demonstrate.
Integrate

Analyze

Break down material into constituent parts

Case studies
Discussions
Questions
Debate
Essays
Presentations
Role play

Assess
Coordinate
Monitor Critique
Conclude
Test Judge

Evaluate

Make judgements based on a set of criteria

Projects
Problems
Case studies
Simulations
Critiques
Debates

Generate Design
Produce Develop
Construct
Formulate

Create

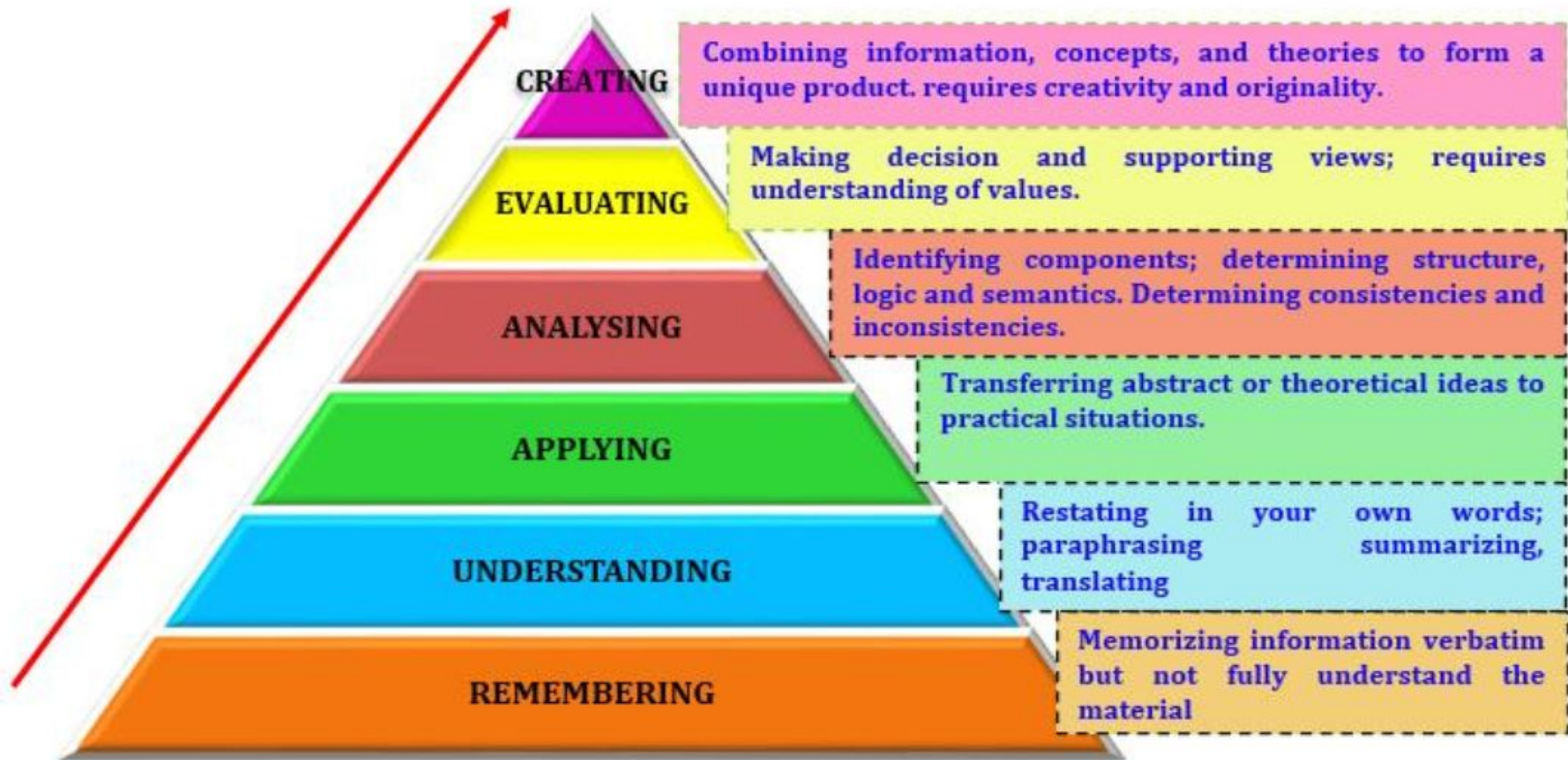
Create an original product, put together separate elements into a coherent whole

Projects
Presentations
Artifacts showcase

Assessments Ideas

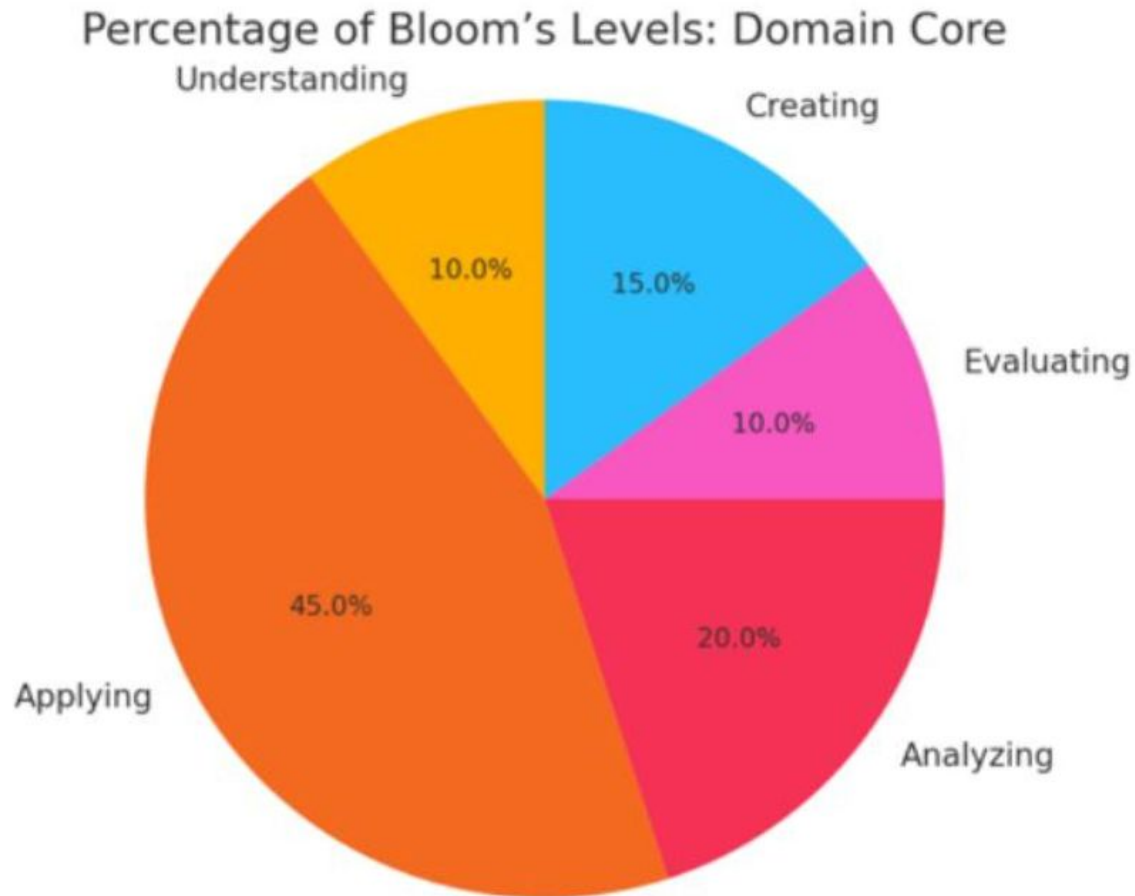
Adapted from [Bloom's Taxonomy Action Verbs and Activities](#) by Lida Hokkanen, licensed under CC BY-NC-SA 4.0.

Cognitive Domain Learning Levels

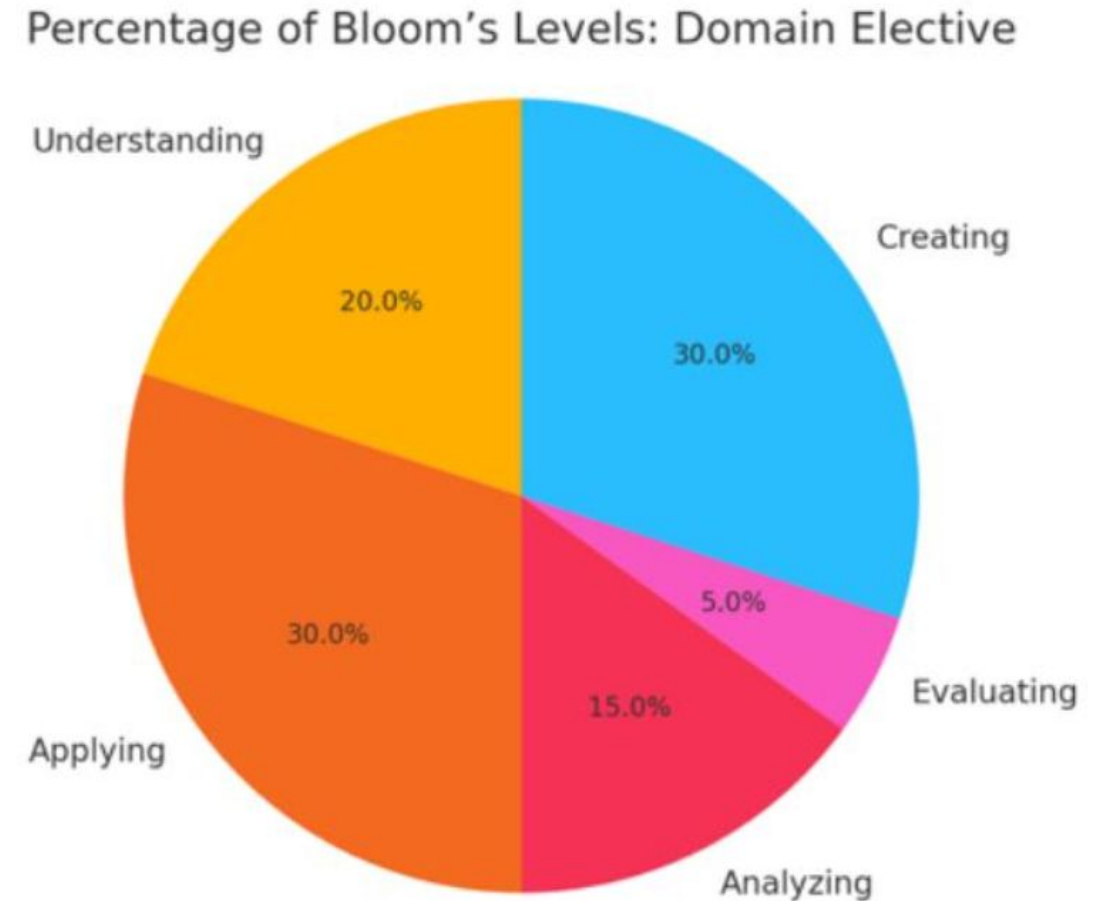


Percentage of Bloom's Levels by Course Category

Domain Core Courses



Domain Elective Courses



Difference between Course Objectives and Course Learning Outcomes

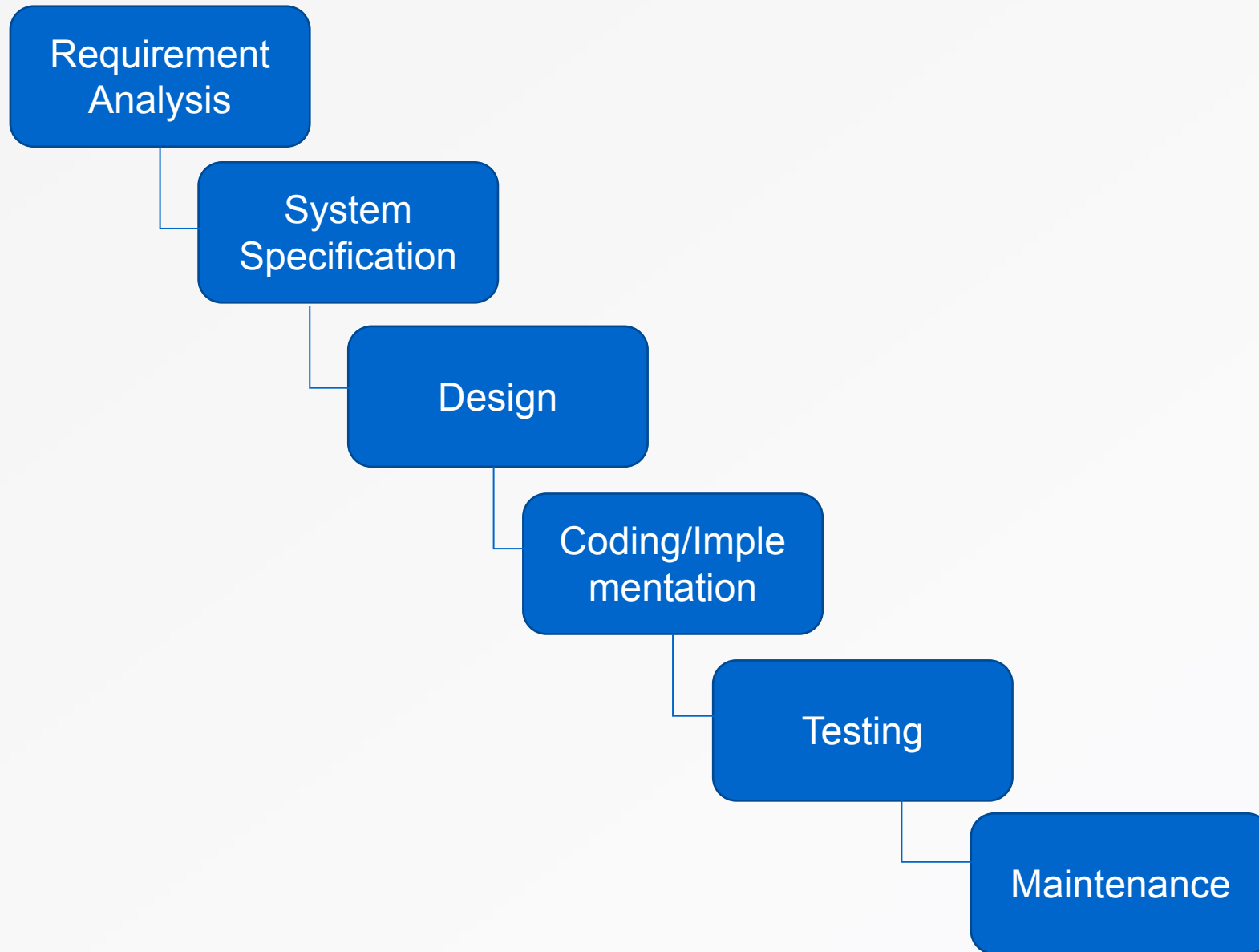
Aspect	Course Objectives	Course Learning Outcomes
Definition	Broad goals of what the course aims to achieve	Specific skills and knowledge students should acquire
Focus	Overall intentions of the course	Measurable student achievements
Nature	General and often qualitative	Specific and often quantitative
Scope	Broad and overarching	Narrow and detailed
Measurement	Difficult to measure directly	Easier to measure through assessments
Examples	Understand the principles of computer science	Explain the basic concepts of algorithms
	Develop problem-solving skills	Implement basic sorting algorithms in a programming language
Audience	Instructors and course designers	Students and instructors
Time Frame	Long-term goals	Achievable within the duration of the course

Review – SEC course

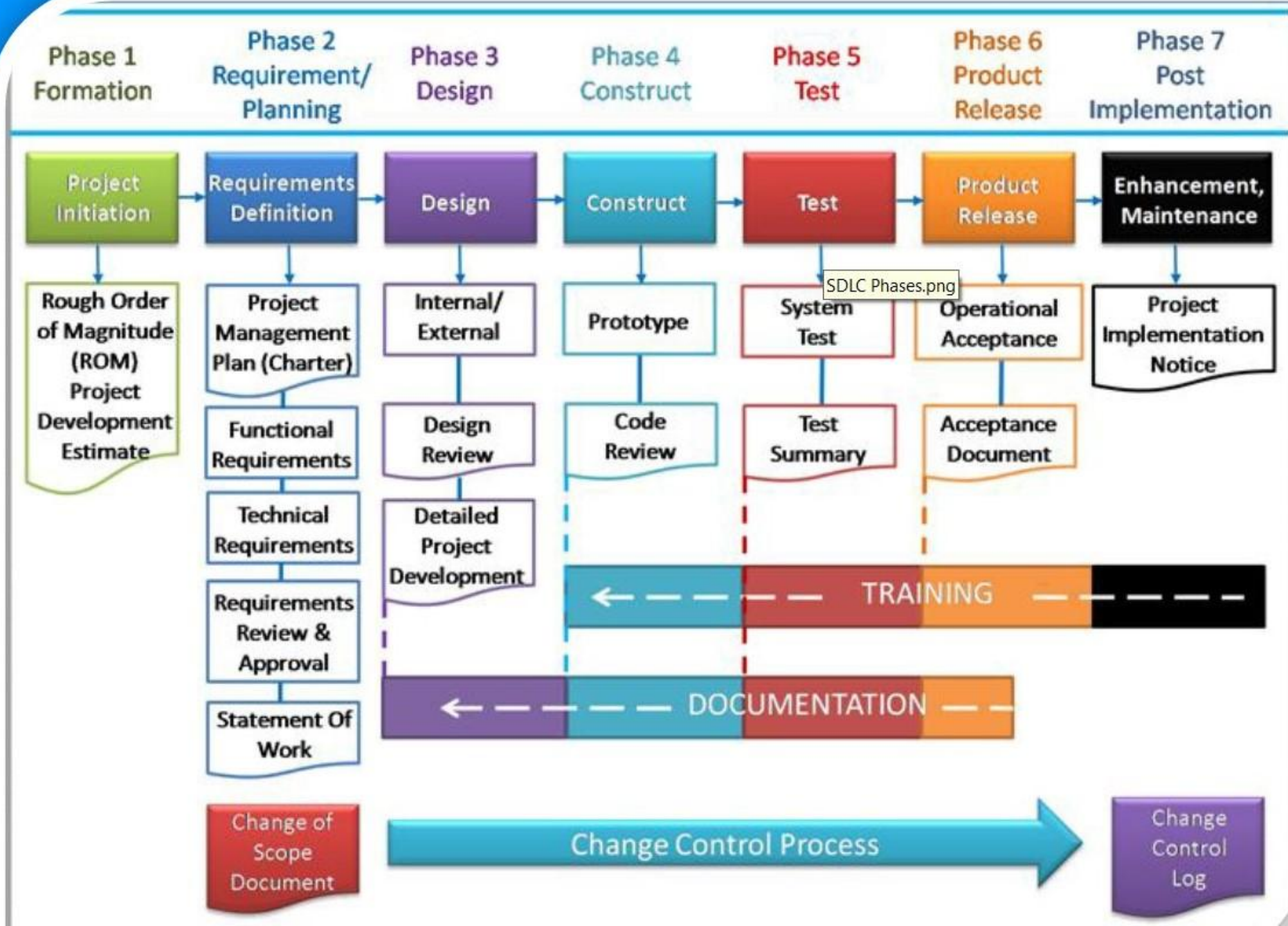
Software Development Process

- Software systems now contain hundreds of **thousands of lines** of code and a much more disciplined engineering approach to software development has had to be adopted.
- Software development involves splitting the development process into a **series of phases**, each phase being associated with a distinct activity.
- These phases are **requirements analysis, system specification, system design, detailed design, coding, integration, operation, and maintenance.**

Software development phases



SDLC



Thanks for your attention!

Any Question?



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